

GAGE, OK, USA

WMO: 746410

Lat: 36.297N

Lon: 99.769W

Elev: 668

StdP: 93.56

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.2	-18.3	0.8	-8.7	-16.2	1.0	-5.9	13.6	10.3	12.4	7.7	3.7	10	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	38.6	21.3	37.1	21.6	35.6	21.5	24.1	32.8	23.5	32.4	22.9	31.9	6.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	18.0	27.3	21.1	17.1	26.6	20.5	16.4	26.1	75.9	32.7	73.3	32.3	71.1	31.9	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	12.0	11.0	-18.0	40.8	2.8	1.7	-20.0	42.1	-21.6	43.1	-23.2	44.0	-25.2	45.3		
(5)			-18.4	25.9	2.7	1.4	-20.3	26.9	-21.9	27.7	-23.5	28.5	-25.4	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	2.3	4.0	9.3	14.3	19.7	24.9	27.3	26.6	22.3	15.2	8.4	2.8		
	DBStd	10.29	5.67	6.33	6.20	5.41	4.78	3.62	3.10	3.14	4.42	5.17	5.49	5.47		
	HDD10.0	874	244	184	87	21	1	0	0	0	0	13	93	231		
	HDD18.3	2291	496	403	285	142	42	2	0	0	15	124	299	483		
	CDD10.0	2634	6	16	66	150	303	447	537	516	369	174	45	6		
	CDD18.3	1011	0	1	5	22	85	199	279	258	134	27	2	0		
	CDH23.3	12389	4	20	106	349	1044	2359	3564	3039	1497	356	50	2		
	CDH26.7	6417	1	5	28	128	467	1210	2040	1692	714	122	9	0		
(14) Wind	WSAvg	5.2	4.7	5.0	5.6	6.2	5.7	5.8	5.0	4.5	4.9	5.1	5.0	4.7		
(15) Precipitation	PrecAvg	487	12	19	38	48	83	71	41	64	42	40	26	18		
	PrecMax	718	36	62	141	153	220	201	95	192	131	200	107	64		
	PrecMin	337	0	0	0	10	0	8	0	0	0	0	0	0		
	PrecStd	86	10	17	35	34	52	43	28	46	30	49	29	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	26.3	29.7	33.5	36.6	39.8	40.7	40.4	38.1	33.1	28.0	22.1		
		MCWB	10.3	11.2	14.1	16.0	19.0	20.1	21.1	21.3	20.6	17.7	13.7	10.3		
	2%	DB	18.8	22.1	26.3	30.4	34.0	36.9	38.6	38.1	35.5	30.0	24.3	18.2		
		MCWB	8.4	10.2	13.5	15.6	19.7	21.4	21.6	21.7	20.6	17.6	12.7	9.4		
	5%	DB	15.3	18.2	23.2	27.4	31.7	34.8	37.2	36.6	33.1	27.4	21.2	15.3		
		MCWB	6.9	9.1	12.6	15.2	19.3	21.6	22.0	21.8	20.5	16.6	12.0	7.4		
	10%	DB	12.3	14.8	20.3	24.5	29.1	32.9	35.5	34.7	31.0	24.7	18.1	12.4		
		MCWB	5.7	7.3	11.3	14.2	18.5	21.7	22.1	21.8	20.0	15.7	10.9	6.6		
	0.4%	WB	11.7	14.2	17.8	19.9	22.9	25.2	25.1	24.7	23.4	21.4	16.7	13.3		
		MCDB	19.8	21.3	24.6	26.6	31.9	32.8	33.4	33.8	32.0	27.5	20.6	17.8		
	2%	WB	9.3	11.6	16.2	18.4	21.7	23.9	24.1	23.7	22.3	19.9	14.9	10.5		
		MCDB	17.2	19.0	21.6	24.5	29.7	32.1	33.0	32.7	30.9	25.2	19.9	16.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	7.6	9.8	14.6	17.0	20.7	23.1	23.5	23.1	21.6	18.7	13.3	8.4		
		MCDB	14.8	18.1	20.2	23.6	28.3	31.6	32.6	32.2	29.9	23.4	19.1	14.8		
	10%	WB	5.9	7.8	12.4	15.5	19.9	22.4	22.9	22.5	20.8	17.2	11.8	6.5		
		MCDB	12.0	14.4	19.6	22.4	26.6	30.5	32.1	31.2	28.7	22.9	17.6	12.2		
	5% DB	MDBR	15.1	15.3	15.7	15.2	14.4	13.5	13.9	14.0	14.3	15.1	15.4	14.3		
		MCDBR	20.5	20.7	19.9	19.2	17.0	15.5	16.0	16.4	16.2	18.2	20.0	19.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	11.7	11.2	9.7	8.5	6.9	5.4	4.4	4.8	5.4	8.2	10.3	11.2		
		MCDBR	18.9	19.1	16.0	15.4	14.7	13.6	13.7	13.7	13.8	13.7	15.8	17.6		
	5% WB	MCWBR	11.2	11.0	8.6	8.2	7.1	5.8	5.0	4.9	5.3	7.4	9.1	10.7		
		MCDBR	11.2	11.0	8.6	8.2	7.1	5.8	5.0	4.9	5.3	7.4	9.1	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.279	0.317	0.342	0.359	0.397	0.400	0.387	0.355	0.327	0.289	0.273		
	taud		2.551	2.538	2.440	2.399	2.423	2.360	2.378	2.372	2.438	2.498	2.544	2.551		
	Ebn at Noon		942	972	953	940	922	883	878	884	899	900	910	916		
	Edh at Noon		76	87	106	117	116	123	120	119	105	90	76	72		
(44) All-Sky Solar Radiation	RadAvg		2.87	3.61	4.79	5.92	6.61	7.23	7.09	6.34	5.36	3.99	3.05	2.49		
	RadStd		0.26	0.43	0.37	0.39	0.58	0.53	0.36	0.34	0.29	0.51	0.33	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+94
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page